Contents

1 Integral Curves and Flows

1.1 Integral Curves

Definition 1.1.1 ▸ Integral Curve

If V is a vector field on M, an ***integral curve of*** V is a differentiable curve $\gamma : J \to M$ whose velocity at each point is equal to the value of V at that point:

$$\gamma'(t) = V_{\gamma(t)} \quad \text{for all } t \in J.$$

If $0 \in J$, the point $\gamma(0)$ is called the ***starting point of*** γ.

Finding integral curves boils down to sloving a system of ordinary differential equations in a smooth chat. Suppose V is a smooth vector field on M and $\gamma : J \to M$ is a smooth curve. On a smooth coordinate domain $U \subseteq M$, we can write γ in local coordinates as

$$\gamma(t) = \left(\gamma^1(t), \cdots, \gamma^n(t)\right)$$

Then the condition $\gamma'(t) = V_{\gamma(t)}$ for γ to be an integral curve of V can be written

$$\dot{\gamma}^i(t) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)} = V^i(\gamma(t)) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)},$$

which reduces to the following autonomous system of ordinary differential equatoins:

$$\begin{aligned} \dot{\gamma}^1(t) &= V^1(\gamma^1(t), \ldots, \gamma^n(t)), \\ &\vdots \\ \dot{\gamma}^n(t) &= V^n(\gamma^1(t), \ldots, \gamma^n(t)). \end{aligned}$$

Proposition 1.1.2

Let V be a smooth vector field on a smooth manifold M. For each point $p \in M$, there exists $\varepsilon > 0$ and a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ that is an integral curve of V starting at p.

Proof. The ODE system corosponding to the condition

$$\gamma'(t) = V_{\gamma(t)}$$

is smooth. Thus it satisfies the local Lipschitz condition. The the proposition follows from the Picard-Lindelöf Theorem. $\square$

1.2 Exercise